

Τμήμα Μηχανικών Ηλεκτρονικών Υπολογιστών & Πληροφορικής

Γραμμική Άλγεβρα (ΕΕΙ_104)

Εργασία Εξαμήνου

ΕΑΡΙΝΟ ΕΞΑΜΗΝΟ 2025

Μαθητής:

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(14/4/2025)

Γραμμική Άλγεβρα - Εργασία Εξαμήνου

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Εx. Set 1.3) 33) $\begin{pmatrix} 4 & 3 & 3 \\ 2 & 1 & 0 \\ 4 & 4 & 2 \end{pmatrix}$ a) $y=\begin{pmatrix} 4 & 3 & 3 \\ 2 & 1 & 0 \\ 4 & 4 & 2 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \\ 7.5 \end{pmatrix}$ b) $y=\begin{pmatrix} 8+3+4\cdot 5 \\ 4+1 \\ 8+4+3 \end{pmatrix}\begin{matrix} 15.5 \\ 5 \\ 15 \end{matrix}$

c) $\begin{pmatrix} 4 & 3 & 3 \\ 2 & 2 & 2 \\ 4 & 4 & 2 \end{pmatrix}=\begin{pmatrix} 2 \\ 1 \\ 7.50 \end{pmatrix}=\begin{pmatrix} 15.5 \\ 9 \\ 15 \end{pmatrix}$

Εξισώσεις:

Εx. Set 1.7) 40a) $Ly=B\begin{bmatrix} -1 & 0 & 0 \\ -2 & 3 & 0 \\ 2 & u & 1 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\begin{bmatrix} 7 \\ -2 \\ 0 \end{bmatrix}$

$\cdot -2y_1+3y_2=-2\cdot 7+3y_2=$
$-2 \gg 7-2 \gg$
$-2+3y_2=-2 \Rightarrow \boxed{y_2=0}$

$2y_1+4y_2+y_3=0 \Rightarrow 62\cdot 7+u(0)+y_3=$
$=0 \Rightarrow 2+y_3=0 \Rightarrow \boxed{y_3=-2}$ $\quad y=\begin{pmatrix} 7 \\ 0 \\ -2 \end{pmatrix}$

$6x=y\cdot\begin{bmatrix} 2 & -7 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$

$\cdot x_3=-2 \Rightarrow x_3=-\frac{7}{2}$
$x_2+2x_3=0 \Rightarrow x_2+2\left(-\frac{7}{2}\right)$
$=0 \Rightarrow x_2-7=0$
$\Rightarrow x_2=7$

$2x_1-x_2+3x_3=7 \Rightarrow 2x_1-7+3\left(-\frac{7}{2}\right)=7 \Rightarrow 2x-7-\frac{3}{2}=7$
$\Rightarrow 2x_1=7+5/2=7/2 \Rightarrow x_1=7/4$ $\quad \boxed{x=\begin{pmatrix} 7/4 \\ 7 \\ -7/2 \end{pmatrix}}$

Εx Set 2.7) 23) $A=\begin{bmatrix} 7 & 7 & u^2 \\ 7 & 7 & u^2 \\ 7 & 7 & u^2 \end{bmatrix}$ $\det(A)=7\begin{vmatrix} u & u^2 \\ u & u^2 \end{vmatrix}u\begin{bmatrix} 7 & n^2 \\ 7 & n^2 \end{bmatrix}n^2\cdot\begin{pmatrix} L & u \\ L & u \end{pmatrix}$

$\hookrightarrow h-n^2-u-n^2=0$ $\hookrightarrow 7\cdot n^2-7\cdot n^2=0$ $\quad 7\cdot 0\cdot 7\cdot u=0$

$\Rightarrow$ αρα $\det(A)=10u\cdot 0+u^2\cdot 0=0$

34) 1) $\begin{vmatrix} 6 & ac \\ e & df \end{vmatrix}=0$ $\quad \begin{bmatrix} a & 6 \\ 0 & c \end{bmatrix}\begin{bmatrix} d & e \\ 0 & f \end{bmatrix}=\begin{bmatrix} ad & ae+6f \\ 0 & cf \end{bmatrix}$

2) $\begin{bmatrix} d & e \\ 0 & f \end{bmatrix}\begin{bmatrix} a & 6 \\ 0 & c \end{bmatrix}=\begin{bmatrix} da & db+ec \\ 0 & fc \end{bmatrix}$ $\quad ad=da$
$ae+6f=df$
$cf=fc$

$ae+bf=db+ec \Rightarrow ae-ec=db-6f$
$\Rightarrow e(a-c)=6(d-f)$
$6(d-f)-e(a-c)=0$ $\begin{bmatrix} 6 & a-e \\ e & d-f \end{bmatrix}=0$

Ex. Set 7.4) 19) $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ a) A^3 ;

$A^2 = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 3\cdot3+7\cdot2 & 3\cdot7+7\cdot7 \\ 2\cdot3+7\cdot2 & 2\cdot7+7\cdot7 \end{bmatrix}$

$= \begin{bmatrix} 9+2 & 3+7 \\ 6+2 & 2+7 \end{bmatrix} = \begin{bmatrix} 11 & 4 \\ 8 & 3 \end{bmatrix}$

$A^2 \cdot A \quad \begin{bmatrix} 11 & 4 \\ 8 & 3 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 11\cdot3+4\cdot2 & 11\cdot7+4\cdot7 \\ 8\cdot3+3\cdot2 & 8\cdot7+3\cdot7 \end{bmatrix}$

$A^3 = \begin{bmatrix} 41 & 15 \\ 30 & 11 \end{bmatrix}$, B) A^{-3}

~~$A^{-1} = \frac{1}{7}\begin{bmatrix} 7 & -7 \\ -2 & 3 \end{bmatrix}\begin{bmatrix} 7 & -7 \\ -2 & 3 \end{bmatrix}$~~

- $A^{-1} = \frac{1}{\det(A)} \cdot adj(A)$, $\det(A) = 3\cdot7 - (2)(7) = 7$
- $A^{-1} = 1/7 \cdot \begin{bmatrix} 7 & -7 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 7 & -7 \\ -2 & 3 \end{bmatrix}$
- $(A^{-1})^2 = \begin{bmatrix} 7 & -7 \\ -2 & 3 \end{bmatrix}\begin{bmatrix} 7 & -7 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 7\cdot7+(-7)(-2) & 7(-7)+(-7)3 \\ -2\cdot7+3(-2) & -2(-7)+3\cdot3 \end{bmatrix} =$

$= \begin{bmatrix} 7+2 & -7-3 \\ -2-6 & 2+9 \end{bmatrix} = \begin{bmatrix} 3 & -4 \\ -8 & 11 \end{bmatrix}$

- $(A^{-1})\cdot A^{-1} = \begin{bmatrix} 3 & -4 \\ -8 & 11 \end{bmatrix}\begin{bmatrix} 7 & -7 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 3\cdot7+(-4)(-2) & 3(-7)+(4)3 \\ -6\cdot7+11-(-2) & -8(-7)+11\cdot3 \end{bmatrix} =$

$= \begin{bmatrix} 3+8 & -3-12 \\ -8\cdot22 & 8+33 \end{bmatrix} = \begin{bmatrix} 11 & -15 \\ -31 & 41 \end{bmatrix}$

- $a^{-3} = \begin{bmatrix} 11 & -15 \\ -30 & 41 \end{bmatrix}$
- $A^2 - 2a - 7$
- $A^2 = \begin{bmatrix} 11 & 4 \\ 8 & 3 \end{bmatrix}$, $2A = 2\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 4 & 2 \end{bmatrix}$, $i = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- $A^2 - 2A - 2 = \begin{bmatrix} 11 & 4 \\ 8 & 3 \end{bmatrix} - \begin{bmatrix} 6 & 2 \\ 4 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 5 & 2 \\ 4 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 4 & 2 \end{bmatrix}$

- $A^2 - 2A + 2 = \begin{bmatrix} 6 & 2 \\ 4 & 2 \end{bmatrix}$

Ex. Set 2.1) (26) Chapter 2

$$A=\begin{bmatrix}4&0&0&7&0\\3&3&0&7&0\\2&3&-7&0&7\\2&4&2&3&0\\2&2&0&0&2\end{bmatrix}$$

Ανάπτυξη πρώτης γραμμής

$\det(A)=a\det(M_{11})+0-0+7a\det(M_{50})+0$

$\det(A)=a\det(M_{11})+2df(M_{77})$

αφαιρούμε την 7η γραμμη και 7η στήλη

$$M_{11}=\begin{bmatrix}3&0&-7&0\\3&-7&0&7\\4&2&3&0\\2&0&0&2\end{bmatrix}$$

⇒ ανάπτυξη 4ης στήλης: $\det(m_{11})=7\cdot(-7)^{4+3}$

$$\det\begin{bmatrix}3&0&7\\3&-7&0\\2&0&0\end{bmatrix}\Rightarrow m$$

$=7(-7)^{5}\det(33)$

$=-\det(33)$

$$\det(m)=7(-7)^{7+3}\begin{vmatrix}3&0\\3&-7\end{vmatrix}=7(7)\cdot3(-7)-0-3$$

$\det(m_{11})=-(-3)=3$ ⇒ αφαιρούμε 7η γραμμη της 4ης στήλης

$$M_{14}=\begin{bmatrix}3&3&0&0\\2&3&-1&7\\2&4&2&0\\2&2&0&2\end{bmatrix}$$

ανάπτυξη 7ης γραμμής

$$\det(M_{14})=3\det\begin{bmatrix}3&-1&7\\4&2&0\\2&0&2\end{bmatrix}3\det\begin{bmatrix}2&-1&7\\2&2&0\\2&0&2\end{bmatrix}$$

$$\Rightarrow\begin{bmatrix}3&-7&7\\4&2&0\\2&0&2\end{bmatrix}$$ ανάπτυξη 3ης στήλης

$$7(-1)^{7+3}\begin{vmatrix}a&2\\2&0\end{vmatrix}=7(7)\cdot(a\cdot a-2\cdot2)=-4$$

$*\Rightarrow\boxed{3(-a)=-12}$

Β, τρόπος:

$$\begin{bmatrix}2&-7&7\\2&2&0\\2&0&2\end{bmatrix}(1)(-1)^{7+3}\begin{vmatrix}2&2\\2&0\end{vmatrix}=7\cdot(2\cdot0-2\cdot7)=-4$$

$(-3)(-4)=+12$

Αρα $\det(A)=4\cdot3+(-12)+(12)=12+0=12\Rightarrow\det A=12$

Ex Set 7.2) 21) $x=\sin\alpha$, $y=\cos\beta$, $z=\tan\gamma$

$$\begin{cases}x+2y+3z=0\\2x+5y+3z=0\\-x-5y+5z=0\end{cases}$$

$$\left(\begin{array}{ccc|c}1&2&3&0\\2&5&3&0\\-1&-5&5&0\end{array}\right)R_2:R_1-2R_1\left(\begin{array}{ccc|c}1&2&3&0\\0&1&-3&0\\0&-3&8&0\end{array}\right)$$

$$R_3:R_3+3R_2\left(\begin{array}{ccc|c}1&2&3&0\\0&-1&-3&0\\0&0&-1&0\end{array}\right)R_3:-R_3\left(\begin{array}{ccc|c}1&2&3&0\\0&-1&-3&0\\0&0&1&0\end{array}\right)$$

$R_1 : R_1 - 3R_3 \begin{pmatrix} 1 & 2 & 0 & | & 0 \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \end{pmatrix}$ $R_1 : R_1 - 2R_2 \begin{pmatrix} 1 & 0 & 0 & | & 0 \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \end{pmatrix}$

$x=0 \Rightarrow \sin\alpha = 0$, $y=0 \Rightarrow \cos\beta = 0$, $z=0 \Rightarrow \tan\gamma = 0$

(22) $\lambda \neq 3$

$y = -(\lambda-5)x$, $y = \frac{-x}{c_\lambda(-3)}$, $c_\lambda(-3) = \frac{1}{c_\lambda - 3}$ $\Rightarrow (c_\lambda - 3)^2 = 1 \Rightarrow (c_\lambda - 3 + 1)(c_\lambda - 3 - 1) = 0$

$\Rightarrow (c_\lambda - 2)(c_\lambda - 4) = 0 \Rightarrow \boxed{c_{\lambda_1} = 2} \; \boxed{c_{\lambda_2} = 4}$

Ex Set 7.5) (7)

a) $[A|I_3] = \begin{bmatrix} 3 & 4 & -7 & | & 1 & 0 & 0 \\ 1 & 0 & 3 & | & 0 & 1 & 0 \\ 2 & 5 & 4 & | & 0 & 0 & 1 \end{bmatrix}$ $R_1 \leftrightarrow R_2$ $\begin{bmatrix} 1 & 0 & 3 & | & 0 & 1 & 0 \\ 3 & 4 & -7 & | & 1 & 0 & 0 \\ 2 & 5 & 4 & | & 0 & 0 & 1 \end{bmatrix}$

$-3R_1$, $-2R_1$ $\begin{bmatrix} 1 & 0 & 3 & | & 0 & 1 & 0 \\ 0 & 4 & -10 & | & 1 & -3 & 0 \\ 0 & 5 & -10 & | & 0 & -2 & 1 \end{bmatrix}$ $-R_3$ $\begin{bmatrix} 1 & 0 & 3 & | & 0 & 1 & 0 \\ 0 & -1 & 0 & | & 1 & -1 & -1 \\ 0 & 5 & -10 & | & 0 & -2 & 1 \end{bmatrix}$

$-R_2$, $+5R_2$ $\begin{bmatrix} 1 & 0 & 3 & | & 0 & 1 & 0 \\ 0 & -1 & 0 & | & -1 & 1 & 1 \\ 0 & 0 & -10 & | & 0 & -2 & 1 \end{bmatrix}$ $\frac{-R_3}{10}$ $\begin{bmatrix} 1 & 0 & 3 & | & 0 & 1 & 0 \\ 0 & 1 & 0 & | & -1 & 1 & 1 \\ 0 & 0 & 1 & | & -0,5 & 0,7 & 0,4 \end{bmatrix}$

$-3R_3$ $\begin{bmatrix} 1 & 0 & 0 & | & 1,5 & -1,7 & -1,2 \\ 0 & 1 & 0 & | & -1 & 1 & 1 \\ 0 & 0 & 1 & | & -0,5 & 0,7 & 0,4 \end{bmatrix}$ $A^{-1} \begin{bmatrix} 1,5 & -1,7 & -1,2 \\ -1 & 1 & 1 \\ -0,5 & 0,7 & 0,4 \end{bmatrix}$

Ex Set 3.7) 21) a) False d) true

b) false e) true

c) false f) false

Ex Set 3.2) 9a)

$u = (3,7,4)$, $v = (2,2,-4)$

$\cdot v \cdot v = 3 \cdot 2 + 7 \cdot 2 + 4(-4)$

$= 6 + 2 - 16 = -8$

$\cdot u \cdot v = 3^2 + 7^2 + 4^2$

$= 4 + 7 + 16 = 26$

$\cdot v \cdot v = 2^2 + 2^2 + (-4)^2$

$= 4 + 4 + 26 = 24$

Ex. Set 5.7) (7) $A = \begin{bmatrix} 4 & 0 & 7 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$

$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$\lambda = 1$, $\lambda = 2$, $\lambda = 3$

για $\lambda = 1$: $\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$, για $\lambda = 2$ $\begin{bmatrix} -7/2 \\ 7 \\ 7 \end{bmatrix}$

για $\lambda = 3$: $\begin{bmatrix} -7 \\ 1 \\ 1 \end{bmatrix}$

10) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ $\lambda^3 - 3\lambda - 2 = 0$

$\boxed{\lambda = 2}$ $\boxed{\lambda = 7 \text{ δρ}}$

για $\lambda = 2$: $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, για $\lambda = -7$: $\begin{bmatrix} -7 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} -7 \\ 0 \\ 1 \end{bmatrix}$

16) $\begin{bmatrix} 2 & -7 & -7 \\ 7 & 0 & -7 \\ -7 & 7 & 2 \end{bmatrix}$ $\lambda^3 - v\lambda^2 + 5\lambda - 2 = 0$

$\lambda = 7$ (δρ) $\lambda = 2$

για $\lambda = 2$: $\begin{bmatrix} -7 \\ -7 \end{bmatrix}$

27) $\begin{pmatrix}1\\-1\\1\end{pmatrix}, \begin{pmatrix}1\\1\\0\end{pmatrix}, \begin{pmatrix}1\\-1\\0\end{pmatrix}$: $A=\begin{bmatrix}1 & 1/2 & -1\\ -1 & 1/2 & 1\\ 1 & 1/2 & 0\end{bmatrix}$ ικανοποιεί: $A_{n\times}^{-1} = \int_1 v_1^{-1}$ για κάθε ιδιοτιμή

Ex. Set) 5.2) (9) a) $\int = 5$ β) $\int = 5$ $\int = 3$

$\int = 3$ $C_5[-A] = 2$ $C_3[-A] = 1$

Ex Set 6.4) 3a) $\{u_1=(1,-1,1), u_2=(2,0,1), u_3=(0,1,1)\}$

$v_1 = \frac{1}{\|w\|}$ $w = \frac{1}{\sqrt{1^2+(-1)^2+1^2}} = \frac{1}{\sqrt{3}} \Rightarrow w = \frac{1}{\sqrt{3}}(1,-1,1)$

$w_2 = u_2 - (u_2 \cdot v_1)v_1 = (2,0,1) - \left[(2,0,1)\frac{1}{\sqrt{3}}(1,-1,1)\right]\cdot\frac{1}{\sqrt{3}}(1,-1,1)$

$= (2,0,1) - \left(\frac{3}{\sqrt{3}}\cdot\frac{1}{\sqrt{3}}(1,-1,1)\right) = (2,0,1) - (1,-1,1) = (1,1,0)$

$v_2 = \frac{1}{\|w_2\|}$, $w_2 = \frac{1}{\sqrt{1^2+1^2+0^2}} \Rightarrow w_2 = \frac{1}{\sqrt{2}}(1,1,0)$

$w_3 = u_3 - (u_3 \cdot v_1)\cdot v_1 - (u_3 \cdot v_2)v_2$

$u_3 \cdot v_1 = (0,1,1)\frac{1}{\sqrt{2}}(1,1,0) = \frac{1}{\sqrt{2}}$

$w_3 = (0,1,1) - (0 \cdot v_1) - \frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}(1,1,0) = \left(-\frac{1}{2},\frac{1}{2},1\right)$

$v_3 = \frac{1}{\|w_3\|}$ $w_3 = \frac{1}{\sqrt{(-\frac{1}{2})^2+(\frac{1}{2})^2+1^2}}\cdot\left(-\frac{1}{2},\frac{1}{2},1\right) = \left(-\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}}\right)$

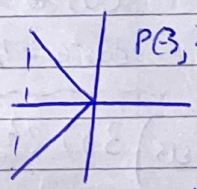
ορθογώνια βάση: $\Rightarrow v_1 = \frac{1}{\sqrt{3}}(1,-1,1)$ $v_2 = \frac{1}{\sqrt{2}}(1,1,0)$

$v_3 = \frac{1}{\sqrt{6}}(-1,1,2)$

Κεφάλαιο 7

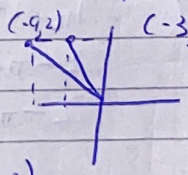
9) a) $P(-3,2)$ T_1 = προβολή προς αξ-x

$M = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} \Rightarrow \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}-3\\2\end{pmatrix} = \begin{pmatrix}-3\\-2\end{pmatrix}$



T_2 = Stretch by a factor of 3 to ox

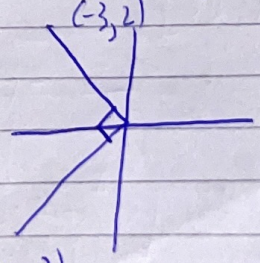
$M = \begin{pmatrix}3 & 0\\ 0 & 1\end{pmatrix} \Rightarrow \begin{pmatrix}3 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}-3\\2\end{pmatrix} = \begin{pmatrix}-9\\2\end{pmatrix}$



T_3 = στροφή 90° από αρχική

$M = \begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix} = \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}$

$\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\begin{pmatrix}-3\\2\end{pmatrix} = \begin{pmatrix}-2\\-3\end{pmatrix}$



Κεφάλαιο 8.2 7α) $A=\begin{pmatrix}1 & -2\\ 4 & 3\end{pmatrix}$ $p(\lambda)=\det(A-\lambda I)$

$$A-\lambda I=\begin{pmatrix}1 & -2\\ 4 & 3\end{pmatrix}-\begin{pmatrix}\lambda & 0\\ 0 & \lambda\end{pmatrix}=\begin{pmatrix}1-\lambda & -2\\ 4 & 3-\lambda\end{pmatrix}$$

$p(\lambda)=\det(A-\lambda I)=(1-\lambda)(3-\lambda)+2\cdot 4=0 \Rightarrow \lambda^2-4\lambda+11=0$

$p(A)=A^2-4A+11=0 \Rightarrow A^2=A\cdot A=\begin{pmatrix}1 & -2\\ 4 & 3\end{pmatrix}\begin{pmatrix}1 & -2\\ 4 & 3\end{pmatrix}=\begin{pmatrix}-7 & -8\\ 16 & 7\end{pmatrix}$

$$p(A)=C_0 I+C_1 A+C_2 A^2=11\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}-4\begin{pmatrix}1 & -2\\ 4 & 3\end{pmatrix}+\begin{pmatrix}-7 & -8\\ 16 & 7\end{pmatrix}=\begin{pmatrix}0 & 0\\ 0 & 0\end{pmatrix}$$

8.3) 1) $2x^2-xy+5y^2,\ x^2+y^2=1$

$A=\begin{pmatrix}2 & -1/2\\ -1/2 & 5\end{pmatrix}$ $\det(A-\lambda I)=(2-\lambda)(5-\lambda)-\frac{1}{4}=0 \Rightarrow 4\lambda^2-28\lambda+39=0$

$\lambda_{1,2}=\frac{28\pm\sqrt{28^2-4\cdot 4\cdot 39}}{2\cdot 4}$ $\lambda_1=\frac{7+\sqrt{10}}{2}$ (max), $\lambda_2=\frac{7-\sqrt{10}}{2}$ (min)

$$[A-\lambda I\,|\,0]=\left[\begin{matrix}\frac{-3-\sqrt{10}}{2} & -1/2\\ -1/2 & \frac{3-\sqrt{10}}{2}\end{matrix}\,\middle|\,\begin{matrix}0\\ 0\end{matrix}\right] \Rightarrow R_2:(3+\sqrt{10})R_2-R_1$$

$R_1: R_1-R_2$

$R_1=-2\cdot R_1$

$R_2=-2R_2$

$3\sqrt{10}$

$$\left[\begin{matrix}1 & 0\\ 0 & 1\end{matrix}\,\middle|\,\begin{matrix}0\\ 0\end{matrix}\right]$$

Κεφάλαιο 7) 11α) $\cos^2\theta=\frac{1}{16}(5\cos\theta+10\cos 3\theta+\cos 5\theta)$

$\cos^2\theta=\frac{1+\cos 2\theta}{2}$, $\cos^4\theta=(\cos^2\theta)^2=\left(\frac{1+\cos^2\theta}{2}\right)^2=\frac{1+2\cos 2\theta+\cos^2 2\theta}{4}$

$\cos^5\theta=\cos\theta-\cos^4\theta=\cos\theta\cdot\frac{1+2\cos\theta+\cos^2 2\theta}{4}$

$\cos^2\theta=1/4,\ \cos+2\cos\theta\cos 2\theta+\cos\theta$

Ασκήσεις 2.3) 3δ) $e^{2-\pi i}=1+i$ $z=(x+yi)$

$\ln e^{2-\pi i}=\ln(1+i)$

$2-\pi i=\ln(1+i)$

$2-\pi i=0$

$x=0$	$\pi=i$

Ασκήσεις 2.4) 7) ii]

$$\int_{\pi/2}^{\pi/2} te^{2it}\,dt = \int_{\pi/4}^{\pi/2} t(\cos 2t + i\sin 2t)\,dt$$

$$= \int_{\pi/4}^{\pi/2} t\cos 2t\,dt + \int_{\pi/4}^{\pi/2} ti\sin 2t\,dt$$

$u = t \Rightarrow du = dt$

$V = \int \cos 2t\,dt$

$\Rightarrow v = \sin 2t/2$

$u = t \Rightarrow du = dt$

$V = \int \sin 2t\,dt$

$\Rightarrow V = -\frac{\cos 2t}{2}$

$$= \left[\frac{t}{2}\sin 2t\right]_{\pi/4}^{\pi/2} - \int_{\pi/4}^{\pi/2} \sin 2t\,dt - \left[\left(\frac{t}{2}\cos 2t\right]_{\pi/4}^{\pi/2} + \int_{\pi/4}^{\pi/2} \cos 2t\,dt\right] i$$

$$= \frac{\pi}{4}\sin\pi - \frac{\pi}{8}\sin\frac{\pi}{2} + \cos\frac{\pi}{4} - \frac{\cos\frac{\pi}{2}}{4} - \left[\frac{\pi}{4}\cos\pi + \frac{\pi}{8}\cos\frac{\pi}{2} + \sin\frac{\pi}{4} - \frac{\sin\pi/2}{4}\right] i$$

$$= \frac{-\pi-2}{8} - i\left(\frac{-\pi-4}{4}\right)$$

Ασκήσεις 3.1)

1a)

$$\int_{7+i}^{7+3i} z^2\,dz = \left[\frac{z^3}{3}\right]_{7+i}^{7+3i} = \frac{(7+3i)^3}{3} - \frac{(7-i)^3}{3}$$

Ασκήσεις 3.2) 2a) $\oint \frac{1}{z^4+4}\,dz$, $\xi = \xi^+$, $(-7+7)$

$$\text{Res}\left[\xi, \sqrt{2}\right] = \lim_{z\to\sqrt{2}} (z+\sqrt{2})\xi(z)$$

$$\oint_{\xi} \frac{1}{z^4+4}\,dz = 2\pi i \sum_{i=1}^{\frac{\pi}{2}} \text{Res}\left[\xi, zi\right]$$

$$\lim_{z\to-\sqrt{2}} \frac{z+\sqrt{2}}{z^4+4} = \frac{2+\sqrt{2}}{-4+4} = \frac{0}{0}$$

$\xi(z) = \frac{1}{z^4+4}$ $\quad z^4+4=0$

$\Rightarrow z^4 = -4$

$\boxed{z = -\sqrt{2}}$